EVALUATION

UUTJ41

Choose the correct answer:

1. The sum of primary valence and secondary valence of the metal M in the complex $\left[M(en)_2(Ox)\right]Cl$ is
   a) 3 b) 6 c) -3 d) 9

2. An excess of silver nitrate is added to 100ml of a 0.01M solution of pentaaquachloridochromium(III)chloride. The number of moles of AgCl precipitated would be
   a)0.02 b) 0.002 c) 0.01 d) 0.2

3. A complex has a molecular formula $MSO_4Cl.6H_2O$ .The aqueous solution of it gives white precipitate with Barium chloride solution and no precipitate is obtained when it is treated with silver nitrate solution. If the secondary valence of the metal is six, which one of the following correctly represents the complex?
   a) $\left[M(H_2O)_4Cl\right]SO_4.2H_2O$ b) $\left[M(H_2O)_6\right]SO_4$
   c) $\left[M(H_2O)_5Cl\right]SO_4.H_2O$ d) $\left[M(H_2O)_3Cl\right]SO_4.3H_2O$

4. Oxidation state of Iron and the charge on the ligand NO in $\left[Fe(H_2O)_5NO\right]SO_4$ are
   a) +2 and 0 respectively b) +3 and 0 respectively
   c) +3 and -1 respectively d) +1 and +1 respectively

5. As per IUPAC guidelines, the name of the complex $\left[Co(en)_2(ONO)Cl\right]Cl$ is
   a) chlorobisethylenediaminenitritocobalt(III) chloride
   b) chloridobis(ethane-1,2-diamine)nitro κ-Ocobaltate(III) chloride
   c) chloridobis(ethane-1,2-diammine)nitrito κ-Ocobalt(II) chloride
   d) chloridobis(ethane-1,2-diammine)nitrito κ -Ocobalt(III)chloride

6. IUPAC name of the complex $K_3\left[Al(C_2O_4)_3\right]$ is
   a) potassiumtrioxalatoaluminium(III)
   b) potassiumtrioxalatoaluminate(II)
   c) potassiumtrisoxalatoaluminate(III)
   d) potassiumtrioxalatoaluminate(III)

XII U5 Coordination jagan.indd 169 04-01-2021 13:24:28